

Licenciatura em Engenharia Informática
43675 – Projeto em Engenharia Informática (2026/2027)
Proposta de projeto

Título

A Backoffice for MariaBike Fleet Intelligence — Fleet Management and personal companion

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Resumo

Build an event-driven backoffice for the mariaBike platform to manage fleet telemetry, individual bike lifecycle, and usage analytics — exposing an extensible service layer for value-added modules such as CO2 footprint tracking, demand forecasting, and BI/ML integration.

Enquadramento e descrição geral

The mariaBike platform (<https://mariabike.com/en/> - [PRR project](#)) already generates a continuous stream of data from embedded sensors, smartwatch connectivity, mobile interfaces, and cloud telemetry. As fleets scale, operators need more than raw data — they need a structured backoffice capable of aggregating, storing, and acting on that data to manage individual bikes, monitor fleet health, and deliver value-added services to riders and operators alike.

Current eBike fleet solutions focus on basic GPS tracking and battery monitoring and lack the architectural foundation to support rich lifecycle management, usage analytics, or extensible data services. Building on the mariaBike platform requires an event-driven backbone that can ingest high-frequency telemetry, maintain per-bike and per-user state, and serve as a clean integration point for downstream modules such as BI dashboards, ML models, or sustainability reporting — without coupling those concerns into the core system.

Objetivos

1. Design and implement an event-driven architecture (e.g. Kafka, MQTT + event store) to ingest and route mariaBike telemetry streams in real time.
2. Build a fleet management layer covering the individual bike lifecycle (registration, maintenance state, usage history) and aggregated fleet metrics (availability, consumption, utilisation).
3. Develop a service layer exposing extensible APIs for value-added services, with at least two concrete implementations (see Candidate Services below).
4. Deliver a back-office dashboard for fleet operators covering telemetry monitoring, bike lifecycle management, and service KPIs.
5. Validate the architecture's extensibility by integrating at least one BI reporting module and one ML-based module (e.g. anomaly detection, demand forecasting).

Possible outcome

A deployed event-driven backoffice for mariaBike fleet management; a service layer with CO₂ footprint and usage analytics modules; a fleet operator dashboard; and an architectural evaluation of extensibility, latency, and scalability under realistic fleet conditions.

Candidate Services

- CO₂ footprint tracking — estimate carbon savings per ride and per user, benchmarked against equivalent car or public transport journeys, and expose aggregated sustainability metrics for operators and riders.
- Usage and rider analytics — trip patterns, energy consumption profiles, and anomaly detection for predictive maintenance across the fleet.
- Demand and availability forecasting — predict bike availability and usage hotspots using historical telemetry, supporting smarter fleet redistribution decisions.
- Rider wellbeing integration — combine fleet telemetry with rider physiological data (from the eBike That Knows Humans proposal) to enrich fleet analytics with safety and fatigue signals.

Lista de atividades (*)

Period	Phase	Description
M1–2	State of the art & design	Fleet management systems, event-driven architectures, mariaBike platform audit.
M3–4	Event backbone	Broker setup, telemetry ingestion pipeline, event schema design.
M5–6	Fleet management layer	Bike lifecycle, usage tracking, fleet metrics API.
M7–8	Service layer	CO ₂ footprint module, usage analytics, extensible API framework.
M9–10	Dashboard & integration	Operator back office UI, BI module, ML module integration.
M11–12	Evaluation	Scalability tests, extensibility assessment

(*) Outcomes and plan may vary depending on the students' background and the definition of the work scope

Observações (apresentar dependências externas (dados, equipamentos, apoio de empresas,...))

The genesis of the proposal is independent of external resources – there are already sample and eBike streams – core contribution from UA in the project - and high-level requirements as a result of the MariaBike project. The UA team involved also DEM/UA (parceiros) and is associated with the Syed Tahir ongoing PhD (funded by FCT Grant)

However, it is not excluded, depending on the evolution of the project and the interest of the MHS (the industry partner), participation in a real-world pilot study (currently under negotiation)